

EMFS and Circuits

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1 Introduction

The test will have the following topics:

- Forces on Parallel Currents
- Toroids and Solenoids
- Magnetic Fields of Moving Charge Particles
- Law of BS
- Current and Torques on Charge Carrying Wires
- Magnetic Forces and Magnet Fundamentals
- Right Hand Rule (RHR)

2 Forces

2.1 Lorentz Force

Forces from a moving particles (Lorentz Forces). Bend $\vec{v}$ into $\vec{b}$ to get the direction for $\vec{F}_B$

$$\vec{F}_B = q(\vec{v} \times \vec{b})$$

Objects moving in a circle.

- $\vec{F}_B = \vec{F}_C$
- $r = \frac{mv}{qb}$

- Pitch: Distance parallel to the magnetic field
- Period of Orbit $v = \frac{2\pi r}{T}$; $T = \frac{2\pi r}{v}$

2.2 Currents and Torques of Wires

$\vec{L}$ is in direction of the current. Field wraps around a wire (RHR it).

$$F_b = i|\vec{L} \times \vec{B}|$$

Torque in a wire loop (probably derivable), with a and b being width and height respectively:

$$\tau = ib(\text{area of loop})$$

3 Law of BS

$$\delta\vec{B} = \frac{\mu_0}{4\pi} \frac{i\delta\vec{L} \times \hat{r}}{r^2}$$

Where $\delta\vec{L}$ is the vectorized length of wire, $\hat{r}$ is the unit vector from $\delta\vec{L}$ to the point, and r is the distance from the point to the wire. Field of a wire on a point:

$$\vec{B} = \frac{\mu_0 i}{2\pi R}$$

Field of an enclosed wire loop:

$$\vec{B} = \frac{\mu_0}{2R}$$

4 Ampere's Law

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{\text{enc}}$$

Amperian Loop: draw a path around an object that mimics the symmetry of the emitter.

5 Magnetic Flux

NOTE: Be able to find flux from relate to velocity, change in area, work, and other forces

1. (Law of Flux) The voltage induced in a coil is equal to negative of the rate of change of magnetic flux through a coil
($\epsilon = -N \frac{\delta\Phi}{\delta t} = -N \frac{\int B \delta A}{t}$)
2. (Lenz's Law) An induced current has a direction such that the magnetic field due to the current opposes the change in the magnetic flux that induces the current.

6 Forces on Parallel Currents

6.1 On Two Wires

$$\left| \frac{F}{L} \right| = \frac{\mu_0 i_1 i_2}{2\pi d}$$

6.2 Wrapped Wires

1. Solenoid: $B = \mu_0 i n$, where the direction is through the solenoid
2. Toroids: $B = \frac{\mu_0 n i}{2\pi r}$

7 Induction

Change of current or change of EMF induces a current in a wire. **Self Inductance** is a type of circuit that creates a field and then tries to change it when the current changes.

1. Sort of like a reverse capacitor
2. Measured in Henries

Inductance of an Inductor

$$L = \frac{\Phi}{i} = \frac{N \int B \cdot \delta A}{i} = \frac{N \int \mu_0 N / L \delta A}{i} = \frac{\mu_0 N^2 A}{L}$$

Induced Voltage of an Inductor:

$$\epsilon = -\frac{\delta \Phi}{\delta t} = -L \frac{\delta i}{\delta t}$$

8 RL Currents

Current Induced in an Inductor over Time

$$\begin{aligned} i(t) &= \frac{\epsilon_0}{R} \left(1 - e^{-\frac{R}{L}t} \right) \\ \epsilon_L(t) &= -L \frac{di}{dt} = -\epsilon_0 e^{-\frac{R}{L}t} \\ V_R(t) &= i(t)R = \epsilon_0 \left(1 - e^{-\frac{R}{L}t} \right) \end{aligned}$$

Energy Stored in a Capacitor

$$U = \frac{1}{2} L i^2$$

Energy Density of a Capacitor

$$u = \frac{B^2}{2\mu_0}$$